DATA SCIENCE PROJECT REPORT

Handwritten Digits Classification using K-Nearest Neighbors (KNN)

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Executive Summary

This project implements a K-Nearest Neighbors (KNN) classifier using scikit-learn's built-in Digits dataset. The workflow includes loading the dataset, preparing a DataFrame, splitting the data into training and testing sets, training the model, evaluating its accuracy, generating a confusion matrix, and producing a classification report.

Project Highlights

Dataset: sklearn Digits Dataset
Algorithm: K-Nearest Neighbors (KNN)
Task: Multi-class Classification
Language: Python
Libraries: Pandas, Scikit-learn, Matplotlib, Seaborn

Introduction

The objective is to classify handwritten digits (0–9) from pixel intensity values. KNN predicts the class of a new sample based on the labels of its nearest neighbors.

Code Explanation

1. Load the Digits dataset using load_digits().
2. Convert the feature matrix into a Pandas DataFrame.
3. Add the target labels as a new column.
4. Split the dataset into training and testing sets using train_test_split().
5. Create a KNeighborsClassifier with n_neighbors=5.
6. Train the model using fit().
7. Evaluate accuracy using score().
8. Predict labels for the test set.
9. Generate a confusion matrix and visualize it with a Seaborn heatmap.
10. Print a detailed classification report including precision, recall, F1-score, and support.

Dataset Overview

The Digits dataset contains grayscale images of handwritten digits represented as 8×8 pixel values. Each sample belongs to one of ten classes (0–9).

Model Evaluation

The notebook evaluates the classifier using test accuracy, a confusion matrix, and a classification report. These metrics measure the classifier's ability to correctly recognize handwritten digits.

Applications

Handwritten digit recognition is widely used in postal mail sorting, bank cheque processing, OCR systems, educational tools, and document digitization.

Future Scope

Improve performance using hyperparameter tuning, cross-validation, weighted KNN, feature scaling, and comparison with algorithms such as SVM, Random Forest, and Neural Networks.

Conclusion

The project demonstrates the implementation of KNN for image classification using a standard benchmark dataset. It provides a strong introduction to supervised machine learning and classification workflows.